

AI in Research

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Some of the terms I could familiarize with:

Jaccard similarity(talks about two sets and their common elements, the ratio of common to uncommon).

Knowledge Graphs structured representation of facts where information is stored as triples in the form of: (Entity 1/sub) _ (Relationship/predicate) _(Entity 2/object).

Embeddings(converts words, sentences and even concepts into vectors(similar words have similar vectors) aiding ML models.

Symbols- Systematic Review Methods based on AL and Snowballing (Active Learning [AI predicts relevant papers based on human reviews, Snowballing starts with AL or relevant papers examines references(backward-sb) and papers that cite them(forward sb)

Language Bias n Framing(positive framing/negative framing), Sugar consumption, ADHD, Global warming case studies.

BiLSTM (Bidirectional Long Short-Term Memory) considers words individually, suitable for small scale text dataset.

GRU (Gated Recurrent Units), help read and understand text in both **forward and backward directions** (ensuring context is captured). Used in Humanities sentiment analysis.

BERT transformer model meaning reads text all at once suitable for largescale tasks(used in visualization summary),self attention feature can relate words even if they are far apart.

CRF (Conditional Random Fields)

What it does: CRF doesn't predict words individually—it considers the entire sequence before finalizing predictions.

Triplet Model is a technique that **compares three things at a time** (two similar papers and one unrelated paper).

The paper **Visual Summary Identification from Scientific Publications** really intrigued me:

It talked about how creating a model which pairs figures in study with inline references to predict a score on whether the content of the figure matches the article's abstract.

One of the major edge it seems to have is instead of treating it as sentence matching, one relates how much the content of the figure caption matches the article's abstract, less text data to scour through.

Here pairs of paragraphs in an article's body and mentioned figures are used as training data making it a self supervised approach that donot necessarily require manually labeled data for training(both time and cost effective).

This approach helps with producing Graphical Abstarcts (GAs, concise, pictorial visual summmmary) in large scale scenarios.

Questions to ponder:

- Is AI data from a Research perspective nuanced enough ?
- If AI is still a tool that focuses more on generating information rather than knowledge is it worth spending time on ?
- Is AI triggering a psychological response in researchers that in-order to produce quality works they should be better at challenging AI answers and prove themselves better ?
- Given the insane amount of papers n studies that has amassed over the years in many research fields is it possible to be attain in depth knowledge over a short span of time in a topic?